

Stochastic

Stochastic sampling is better than greedy sampling.

Stochastic creates a balance between 2 things that is exploration and exploitation . In exploration which means that method allows us to be able to explore new paths and new cities and it searches for new information all the time. When this happens it might find paths that are bad and it might lead to a late convergence or convergence will be much slow.

Stochastic balances this and for exploitation , exploitation means I have some knowledge about the cities and path and I am trying to obtain or retrieve the best solution out of it. We are not exploring anything new here so it might get stuck in a local optima value. If we would have explored more paths we would have found better paths and obtained better results.

Small N

Travelling Salesman Problem using Ant Colony Optimization

100 99 8

100 29

35 99

3 71

26 10

53 78

32 28

71 37

64 18

Hyperparameters

T = 20

InitialPheromone = 0.8

m = 8

epsilon = 1e-06

rho = 0.3

beta = 1.0

alpha = 1.0

Q = 1

Iteration 1: Best distance Obtained = 292.3545728981788

Iteration 2: Best distance Obtained = 292.35457289817873

Iteration 3: Best distance Obtained = 292.35457289817873

Iteration 4: Best distance Obtained = 292.3545728981787

Iteration 5: Best distance Obtained = 292.3545728981787

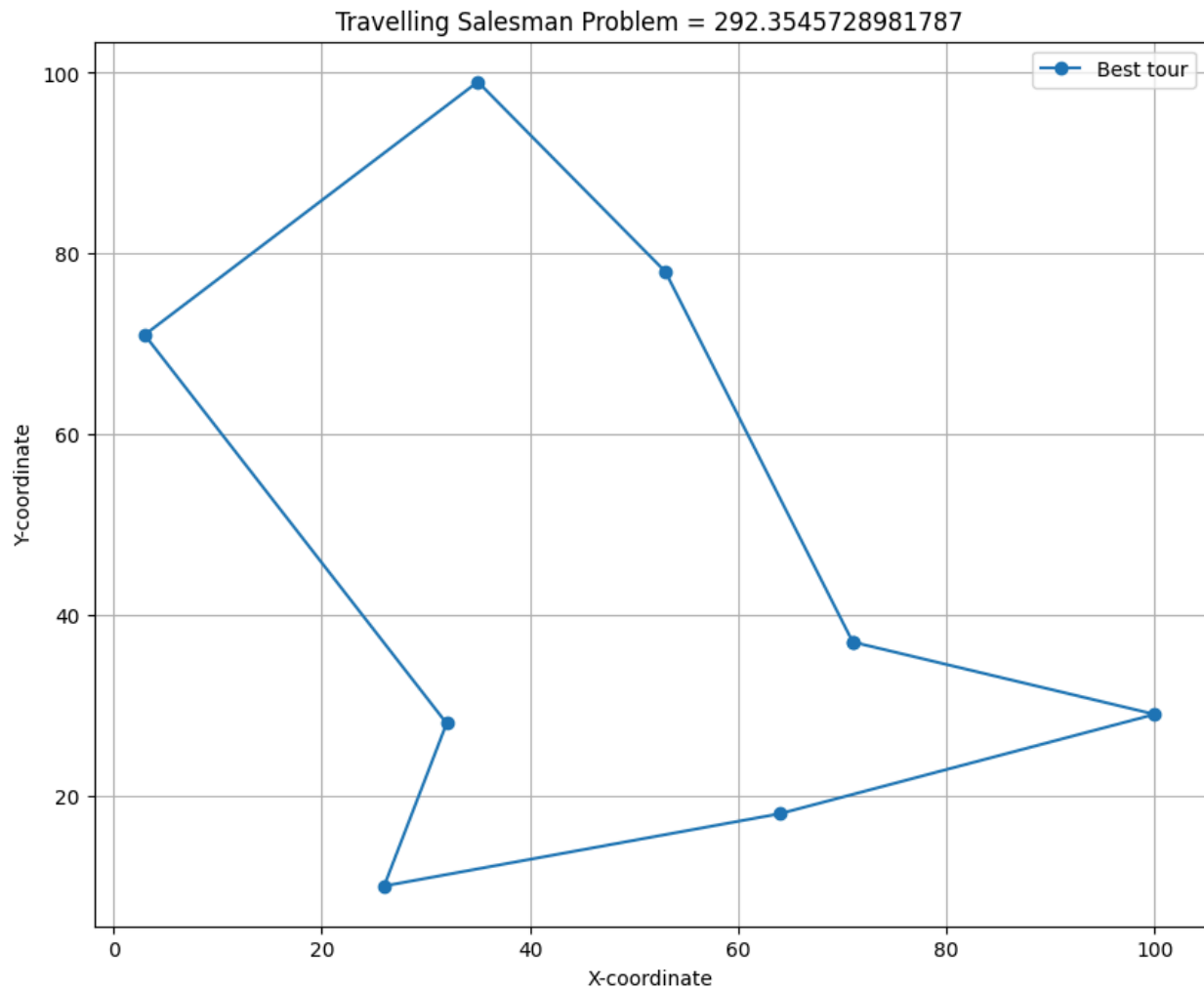
Iteration 6: Best distance Obtained = 292.3545728981787

Iteration 7: Best distance Obtained = 292.3545728981787

Iteration 8: Best distance Obtained = 292.3545728981787

Iteration 9: Best distance Obtained = 292.3545728981787

Iteration 10: Best distance Obtained = 292.3545728981787
Iteration 11: Best distance Obtained = 292.3545728981787
Iteration 12: Best distance Obtained = 292.3545728981787
Iteration 13: Best distance Obtained = 292.3545728981787
Iteration 14: Best distance Obtained = 292.3545728981787
Iteration 15: Best distance Obtained = 292.3545728981787
Iteration 16: Best distance Obtained = 292.3545728981787
Iteration 17: Best distance Obtained = 292.3545728981787
Iteration 18: Best distance Obtained = 292.3545728981787
Iteration 19: Best distance Obtained = 292.3545728981787
Iteration 20: Best distance Obtained = 292.3545728981787
Best Path =
6 0 7 3 5 2 1 4 6
Best Path Length = 292.3545728981787



Large N

Travelling Salesman Problem using Ant Colony Optimization

800 600 60

791 467

51 417

253 293

372 528

634 47

251 491

115 68

378 458

657 174

209 180

642 556

328 215

327 93

78 416

114 190

58 202

172 57

702 461

589 225

424 26

70 575

43 428

286 585

711 173

554 295

203 73

88 596

290 331

547 568

310 320

522 13

222 568

381 46

671 276

303 570

145 48

76 573

110 341

514 162

409 221

46 268

217 212

695 335

455 107

427 194

207 565

686 101

748 381
457 467
790 171
19 360
685 487
228 95
9 147
721 143
563 570
582 167
102 180
39 229
23 309

Hyperparameters

T = 20
InitialPheromone = 0.3
m = 55
epsilon = 1e-06
rho = 0.3
beta = 1.0
alpha = 1.0
Q = 10

Iteration 1: Best distance Obtained = 4112.2415991977405
Iteration 2: Best distance Obtained = 4050.9391454113866
Iteration 3: Best distance Obtained = 4050.9391454113866
Iteration 4: Best distance Obtained = 4050.9391454113866
Iteration 5: Best distance Obtained = 4050.9391454113866
Iteration 6: Best distance Obtained = 4050.9391454113866
Iteration 7: Best distance Obtained = 4050.9391454113866
Iteration 8: Best distance Obtained = 4050.9391454113866
Iteration 9: Best distance Obtained = 4050.9391454113866
Iteration 10: Best distance Obtained = 4050.9391454113866
Iteration 11: Best distance Obtained = 4050.9391454113866
Iteration 12: Best distance Obtained = 4050.93914541137
Iteration 13: Best distance Obtained = 4050.93914541137
Iteration 14: Best distance Obtained = 4050.93914541137
Iteration 15: Best distance Obtained = 4050.93914541137
Iteration 16: Best distance Obtained = 4050.93914541137
Iteration 17: Best distance Obtained = 4050.93914541137
Iteration 18: Best distance Obtained = 4050.93914541137
Iteration 19: Best distance Obtained = 4050.93914541137
Iteration 20: Best distance Obtained = 4050.93914541137

Best Path =

32 19 30 4 46 54 49 23 8 56 18 24 33 42 47 0 17 51 10 55 28 48 7 3 34 22 5 31 45 26 36 20 21 1 13 37
50 59 40 58 15 14 57 53 6 35 16 25 52 9 41 2 27 29 11 39 44 38 43 12 32

Best Path Length = 4050.93914541137

Travelling Salesman Problem = 4050.93914541137

